

DEEPAKRAJ B

ERPNext & Frappe Developer | Backend & Manufacturing Systems

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PROFILE

Software Engineer with hands-on experience designing, developing, and deploying **ERPNext & Frappe Framework** manufacturing platforms for enterprise operations. Specialized in custom module engineering, DocType lifecycle logic, multi-warehouse inventory systems, and shop-floor digitization for a 700-employee manufacturing enterprise across 4 production units. Proficient in **Python, REST API design, Webhooks, MySQL query optimization**, Android scanner integrations, and Dockerized cloud deployments. Experienced in owning full implementation cycles from requirement analysis through production support.

PROFESSIONAL EXPERIENCE

Software Engineer | Kritilabs Technologies Pvt. Ltd., Chennai, India

July 2024 - Present

ERPNext Manufacturing Platform: Digital Lock Manufacturing **Frappe, ERPNext, Python, MySQL, Docker, REST APIs**

- Delivered end-to-end implementation of an **ERPNext / Frappe** manufacturing platform for a 700-employee digital lock manufacturer operating across 4 production units, managing requirements, design, testing, and go-live deployment.
- Engineered custom **Frappe DocTypes, Server Scripts, and Client Scripts** for Bill of Materials (BOM) management, production planning, multi-location inventory tracking, and batch/serial traceability.
- Developed real-time shop-floor dashboards for live assembly line visibility and integrated **Android mobile scanners** via Frappe REST APIs for barcode-based production logging, reducing manual entry by **80%**.
- Built custom **REST APIs and Webhook pipelines** bridging legacy ERP, inventory, and QA systems to ensure reliable cross-system data synchronization without manual intervention.
- Optimized **MySQL** schemas, composite indexes, and transaction queries for high-frequency inventory movements, and containerized the platform using **Docker** with automated **CI/CD deployment pipelines**.

Enterprise Systems, AI Query Tools & Backend Services

Python, FastAPI, LangGraph, PostgreSQL, Docker, AWS

- Engineered an **NL2SQL Query Engine** using **Python, FastAPI, and LangGraph** to translate natural language prompts into verified SQL queries with automated schema validation and execution guardrails.
- Architected backend data ingestion pipelines and tracking services for enterprise IoT systems deployed across **Indian Railways** and logistics monitoring platforms for **IOCL and BPCL**.
- Implemented asynchronous REST endpoints and caching layers (**Redis**) to handle concurrent telemetry streams and reduce reporting latency across distributed services.

TECHNICAL SKILLS

ERP & Frameworks: ERPNext, Frappe Framework, DocTypes, Server/Client Scripts

Backend & Languages: Python (3.8+), FastAPI, SQL, RESTful APIs, Webhooks, JSON

Integration & Frontend: Android Barcode Integration, JavaScript, Jinja2, HTML5/CSS3

AI & Query Tooling: LangGraph, LangChain, NL2SQL, Vector DBs, ChromaDB

Databases & Optimization: MySQL, MariaDB, PostgreSQL, Redis, Schema Indexing

Manufacturing Operations: BOM Management, Work Orders, Inventory, Serial Tracking

Cloud & DevOps: Docker, Docker Compose, Linux, AWS (EC2, S3), CI/CD, Git

Testing & Collaboration: Pytest, Postman, Bench CLI, Nginx, JIRA, Agile/Scrum

EDUCATION

B.Tech in Computer Science & Engineering | SASTRA Deemed University, Thanjavur

CGPA: 7.2955 | 2020 - 2024

CERTIFICATIONS

- Python Crash Course - Google
- Backend Development Specialization - Meta
- Software Engineering Specialization - IBM
- Database Systems & SQL Optimization